

Class 10: Halloween Project

Barry

Table of contents

Background	1
Data Import	1
Data Exploration	2
Winpercent vs Pricepercent	15
Correlation	16
Principal Component Analysis	17

Background

It is Halloween today and to help get in the mood for tonight let's explore some candy preference data from 538.

Data Import

```
candy <- read.csv("candy-data.csv", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860
					66.97173

3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Data Exploration

```
flextable::flextable( head(candy) )
```

chocolate	fruity	caramel	peanut	almond	nougat	crisp	edible	wafer	hard	bar	pluribus s
1	0	1	0	0	0	1	0	0	1	0	0
1	0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	1	0	0

```
library(dplyr)
candy |> nrow()
```

```
[1] 85
```

```
nrow(candy)
```

```
[1] 85
```

```
round( mean(candy$winpercent), 4)
```

```
[1] 50.3168
```

```
candy |> select(winpercent)
```

	winpercent
100 Grand	66.97173
3 Musketeers	67.60294
One dime	32.26109
One quarter	46.11650
Air Heads	52.34146
Almond Joy	50.34755
Baby Ruth	56.91455
Boston Baked Beans	23.41782
Candy Corn	38.01096
Caramel Apple Pops	34.51768
Charleston Chew	38.97504
Chewey Lemonhead Fruit Mix	36.01763
Chiclets	24.52499
Dots	42.27208
Dum Dums	39.46056
Fruit Chews	43.08892
Fun Dip	39.18550
Gobstopper	46.78335
Haribo Gold Bears	57.11974
Haribo Happy Cola	34.15896
Haribo Sour Bears	51.41243
Haribo Twin Snakes	42.17877
Hershey's Kisses	55.37545
Hershey's Krackel	62.28448
Hershey's Milk Chocolate	56.49050
Hershey's Special Dark	59.23612
Jawbusters	28.12744
Junior Mints	57.21925
Kit Kat	76.76860
Laffy Taffy	41.38956
Lemonhead	39.14106
Lifesavers big ring gummies	52.91139
Peanut butter M&M's	71.46505
M&M's	66.57458
Mike & Ike	46.41172
Milk Duds	55.06407
Milky Way	73.09956
Milky Way Midnight	60.80070
Milky Way Simply Caramel	64.35334

Mounds	47.82975
Mr Good Bar	54.52645
Nerds	55.35405
Nestle Butterfinger	70.73564
Nestle Crunch	66.47068
Nik L Nip	22.44534
Now & Later	39.44680
Payday	46.29660
Peanut M&Ms	69.48379
Pixie Sticks	37.72234
Pop Rocks	41.26551
Red vines	37.34852
Reese's Miniatures	81.86626
Reese's Peanut Butter cup	84.18029
Reese's pieces	73.43499
Reese's stuffed with pieces	72.88790
Ring pop	35.29076
Rolo	65.71629
Root Beer Barrels	29.70369
Runts	42.84914
Sixlets	34.72200
Skittles original	63.08514
Skittles wildberry	55.10370
Nestle Smarties	37.88719
Smarties candy	45.99583
Snickers	76.67378
Snickers Crisper	59.52925
Sour Patch Kids	59.86400
Sour Patch Tricksters	52.82595
Starburst	67.03763
Strawberry bon bons	34.57899
Sugar Babies	33.43755
Sugar Daddy	32.23100
Super Bubble	27.30386
Swedish Fish	54.86111
Tootsie Pop	48.98265
Tootsie Roll Juniors	43.06890
Tootsie Roll Midgies	45.73675
Tootsie Roll Snack Bars	49.65350
Trolli Sour Bites	47.17323
Twix	81.64291
Twizzlers	45.46628
Warheads	39.01190

Welch's Fruit Snacks	44.37552
Werther's Original Caramel	41.90431
Whoppers	49.52411

```
win <- candy$winpercent
win.mean <- mean(win)
round(win.mean)
```

```
[1] 50
```

```
candy %>% select(winpercent)
```

	winpercent
100 Grand	66.97173
3 Musketeers	67.60294
One dime	32.26109
One quarter	46.11650
Air Heads	52.34146
Almond Joy	50.34755
Baby Ruth	56.91455
Boston Baked Beans	23.41782
Candy Corn	38.01096
Caramel Apple Pops	34.51768
Charleston Chew	38.97504
Chewey Lemonhead Fruit Mix	36.01763
Chiclets	24.52499
Dots	42.27208
Dum Dums	39.46056
Fruit Chews	43.08892
Fun Dip	39.18550
Gobstopper	46.78335
Haribo Gold Bears	57.11974
Haribo Happy Cola	34.15896
Haribo Sour Bears	51.41243
Haribo Twin Snakes	42.17877
Hershey's Kisses	55.37545
Hershey's Krackel	62.28448
Hershey's Milk Chocolate	56.49050
Hershey's Special Dark	59.23612
Jawbusters	28.12744
Junior Mints	57.21925

Kit Kat	76.76860
Laffy Taffy	41.38956
Lemonhead	39.14106
Lifesavers big ring gummies	52.91139
Peanut butter M&M's	71.46505
M&M's	66.57458
Mike & Ike	46.41172
Milk Duds	55.06407
Milky Way	73.09956
Milky Way Midnight	60.80070
Milky Way Simply Caramel	64.35334
Mounds	47.82975
Mr Good Bar	54.52645
Nerds	55.35405
Nestle Butterfinger	70.73564
Nestle Crunch	66.47068
Nik L Nip	22.44534
Now & Later	39.44680
Payday	46.29660
Peanut M&Ms	69.48379
Pixie Sticks	37.72234
Pop Rocks	41.26551
Red vines	37.34852
Reese's Miniatures	81.86626
Reese's Peanut Butter cup	84.18029
Reese's pieces	73.43499
Reese's stuffed with pieces	72.88790
Ring pop	35.29076
Rolo	65.71629
Root Beer Barrels	29.70369
Runts	42.84914
Sixlets	34.72200
Skittles original	63.08514
Skittles wildberry	55.10370
Nestle Smarties	37.88719
Smarties candy	45.99583
Snickers	76.67378
Snickers Crisper	59.52925
Sour Patch Kids	59.86400
Sour Patch Tricksters	52.82595
Starburst	67.03763
Strawberry bon bons	34.57899
Sugar Babies	33.43755

Sugar Daddy	32.23100
Super Bubble	27.30386
Swedish Fish	54.86111
Tootsie Pop	48.98265
Tootsie Roll Juniors	43.06890
Tootsie Roll Midgies	45.73675
Tootsie Roll Snack Bars	49.65350
Trolli Sour Bites	47.17323
Twix	81.64291
Twizzlers	45.46628
Warheads	39.01190
Welch's Fruit Snacks	44.37552
Werther's Original Caramel	41.90431
Whoppers	49.52411

```
candy |>
  select(winpercent)
```

	winpercent
100 Grand	66.97173
3 Musketeers	67.60294
One dime	32.26109
One quarter	46.11650
Air Heads	52.34146
Almond Joy	50.34755
Baby Ruth	56.91455
Boston Baked Beans	23.41782
Candy Corn	38.01096
Caramel Apple Pops	34.51768
Charleston Chew	38.97504
Chewey Lemonhead Fruit Mix	36.01763
Chiclets	24.52499
Dots	42.27208
Dum Dums	39.46056
Fruit Chews	43.08892
Fun Dip	39.18550
Gobstopper	46.78335
Haribo Gold Bears	57.11974
Haribo Happy Cola	34.15896
Haribo Sour Bears	51.41243
Haribo Twin Snakes	42.17877
Hershey's Kisses	55.37545

Hershey's Krackel	62.28448
Hershey's Milk Chocolate	56.49050
Hershey's Special Dark	59.23612
Jawbusters	28.12744
Junior Mints	57.21925
Kit Kat	76.76860
Laffy Taffy	41.38956
Lemonhead	39.14106
Lifesavers big ring gummies	52.91139
Peanut butter M&M's	71.46505
M&M's	66.57458
Mike & Ike	46.41172
Milk Duds	55.06407
Milky Way	73.09956
Milky Way Midnight	60.80070
Milky Way Simply Caramel	64.35334
Mounds	47.82975
Mr Good Bar	54.52645
Nerds	55.35405
Nestle Butterfinger	70.73564
Nestle Crunch	66.47068
Nik L Nip	22.44534
Now & Later	39.44680
Payday	46.29660
Peanut M&Ms	69.48379
Pixie Sticks	37.72234
Pop Rocks	41.26551
Red vines	37.34852
Reese's Miniatures	81.86626
Reese's Peanut Butter cup	84.18029
Reese's pieces	73.43499
Reese's stuffed with pieces	72.88790
Ring pop	35.29076
Rolo	65.71629
Root Beer Barrels	29.70369
Runts	42.84914
Sixlets	34.72200
Skittles original	63.08514
Skittles wildberry	55.10370
Nestle Smarties	37.88719
Smarties candy	45.99583
Snickers	76.67378
Snickers Crisper	59.52925

Sour Patch Kids	59.86400
Sour Patch Tricksters	52.82595
Starburst	67.03763
Strawberry bon bons	34.57899
Sugar Babies	33.43755
Sugar Daddy	32.23100
Super Bubble	27.30386
Swedish Fish	54.86111
Tootsie Pop	48.98265
Tootsie Roll Juniors	43.06890
Tootsie Roll Midgies	45.73675
Tootsie Roll Snack Bars	49.65350
Trolli Sour Bites	47.17323
Twix	81.64291
Twizzlers	45.46628
Warheads	39.01190
Welch's Fruit Snacks	44.37552
Werther's Original Caramel	41.90431
Whoppers	49.52411

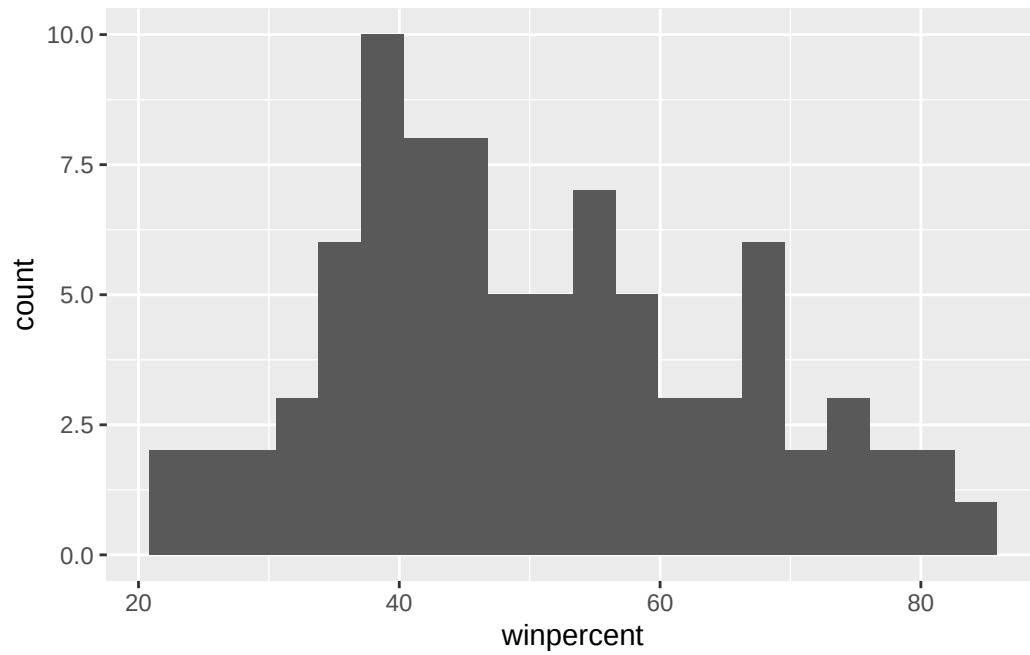
```
colnames(candy)
```

[1] "chocolate"	"fruity"	"caramel"	"peanutyalmondy"
[5] "nougat"	"crispedricewafer"	"hard"	"bar"
[9] "pluribus"	"sugarpercent"	"pricepercent"	"winpercent"

Q8. Plot a histogram of winpercent values

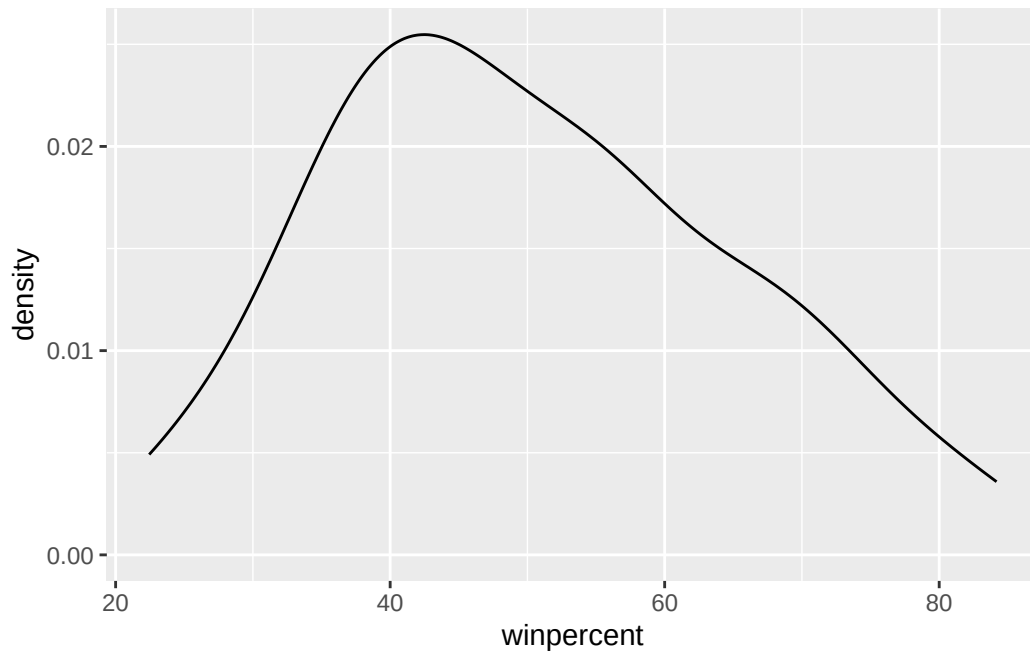
```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

```
ggplot(candy) +  
  aes(winpercent) +  
  geom_density()
```



Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# 1. Find all chocolate candy in the dataset
choc.inds <- as.logical(candy$chocolate)
choc.candy <- candy[choc.inds,]

# 2. Extract their `winpercent` values
choc.win <- choc.candy$winpercent

# 3. Find the mean of these values
choc.mean <- mean(choc.win)
```

```
# 4-6. Do the same for fruity candy
fruit.win <- candy[candy$fruity==1,]$winpercent
fruit.mean <- mean(fruit.win)

# 7. Which mean value is higher
choc.mean
```

```
[1] 60.92153
```

```
fruit.mean
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Q13. What are the five least liked candy types in this set?

```
x <- c(10, 2, 5, 1)
order(x)
```

```
[1] 4 2 3 1
```

```
ord.ind <- order(candy$winpercent)
head( candy[ord.ind, ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197	0.976
Boston Baked Beans				0	0	0	1	0.313	0.511
Chiclets				0	0	0	1	0.046	0.325
Super Bubble				0	0	0	0	0.162	0.116
Jawbusters				0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
tail( candy[ord.ind, ], 5)
```

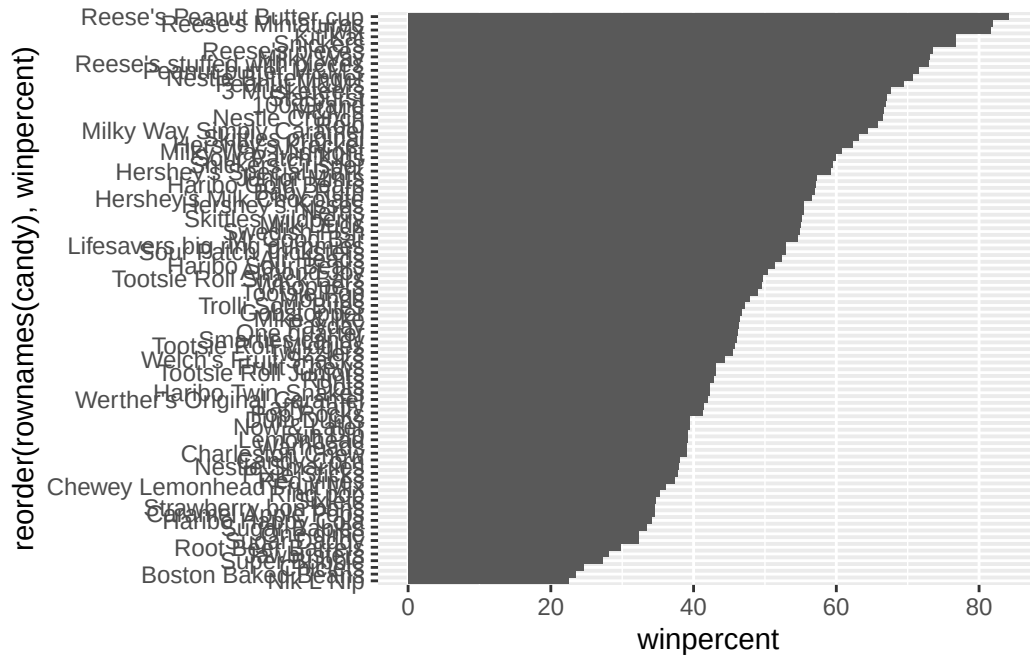
	chocolate	fruity	caramel	peanut	almond	nougat
Snickers	1	0	1		1	1
Kit Kat	1	0	0		0	0
Twix	1	0	1		0	0
Reese's Miniatures	1	0	0		1	0
Reese's Peanut Butter cup	1	0	0		1	0

	crisped	rice	wafer	hard bar	pluribus	sugar	percent
Snickers				0	0	1	0.546
Kit Kat				1	0	1	0.313
Twix				1	0	1	0.546
Reese's Miniatures				0	0	0	0.034
Reese's Peanut Butter cup				0	0	0	0.720

	price	percent	winpercent
Snickers	0.651		76.67378
Kit Kat	0.511		76.76860
Twix	0.906		81.64291
Reese's Miniatures	0.279		81.86626
Reese's Peanut Butter cup	0.651		84.18029

Q15. Make a first barplot of candy ranking based on winpercent values.

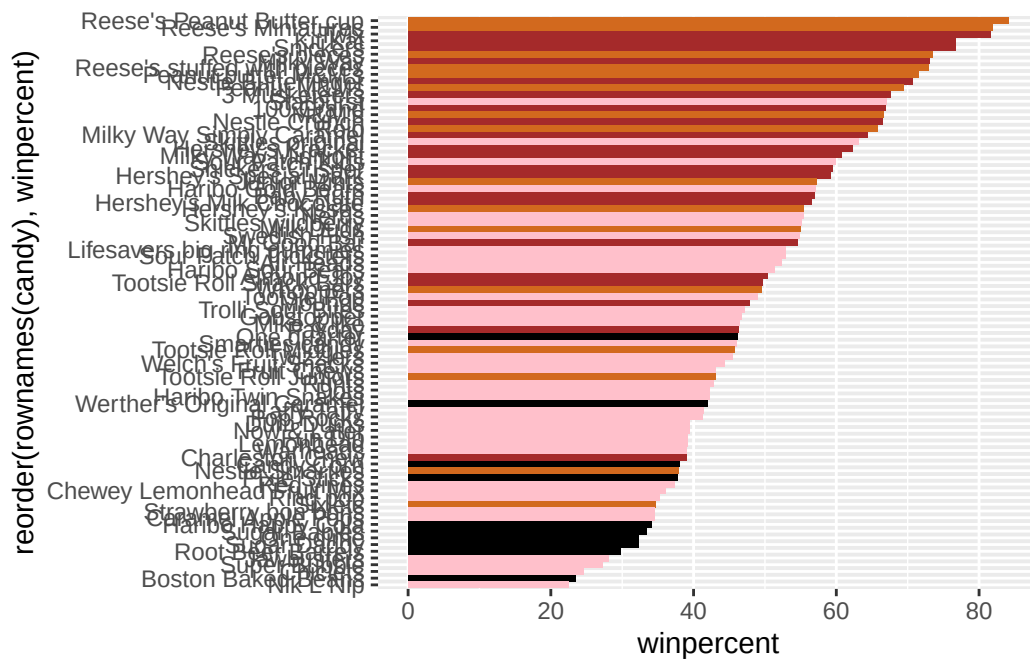
```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



Add some color

```
my_cols <- rep("black", nrow(candy))
my_cols[candy$chocolate==1] <- "chocolate"
my_cols[candy$bar==1] <- "brown"
my_cols[candy$fruity==1] <- "pink"
#my_cols

ggplot(candy) +
  aes(x=winpercent,
      y=reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols)
```



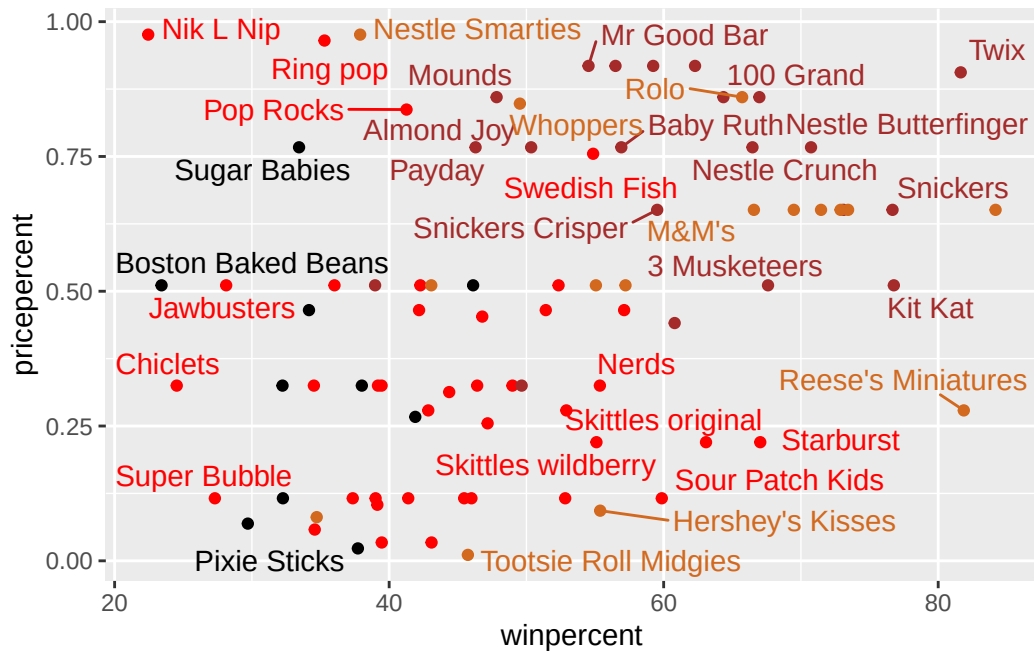
Winpercent vs Pricepercent

```
library(ggrepel)

my_cols[candy$fruity==1] <- "red"

ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols)
```

Warning: ggrepel: 50 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Correlation

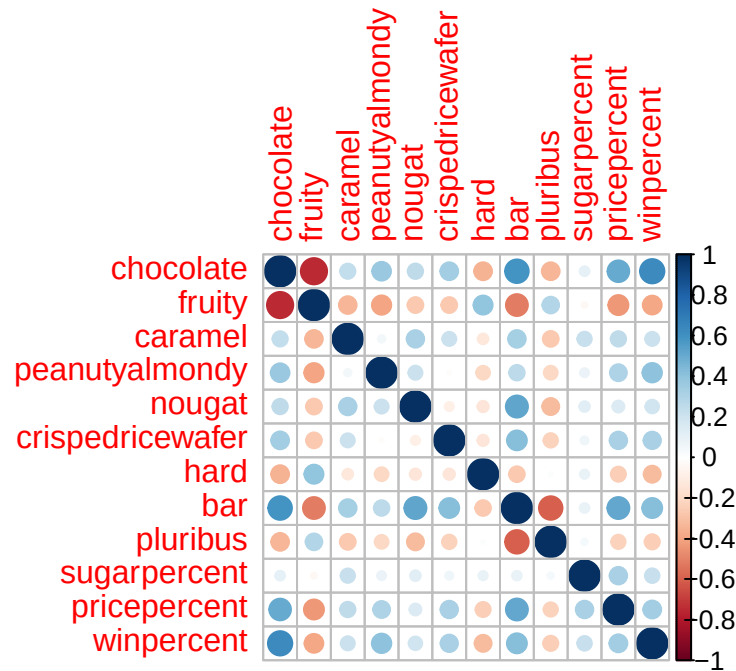
```
cij <- cor(candy)
```

```
library(corrplot)
```

```
corrplot 0.95 loaded
```



```
corrplot(cij)
```



Principal Component Analysis

The main function in base R for this `prcomp()` and we want to set `scale=TRUE` here:

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

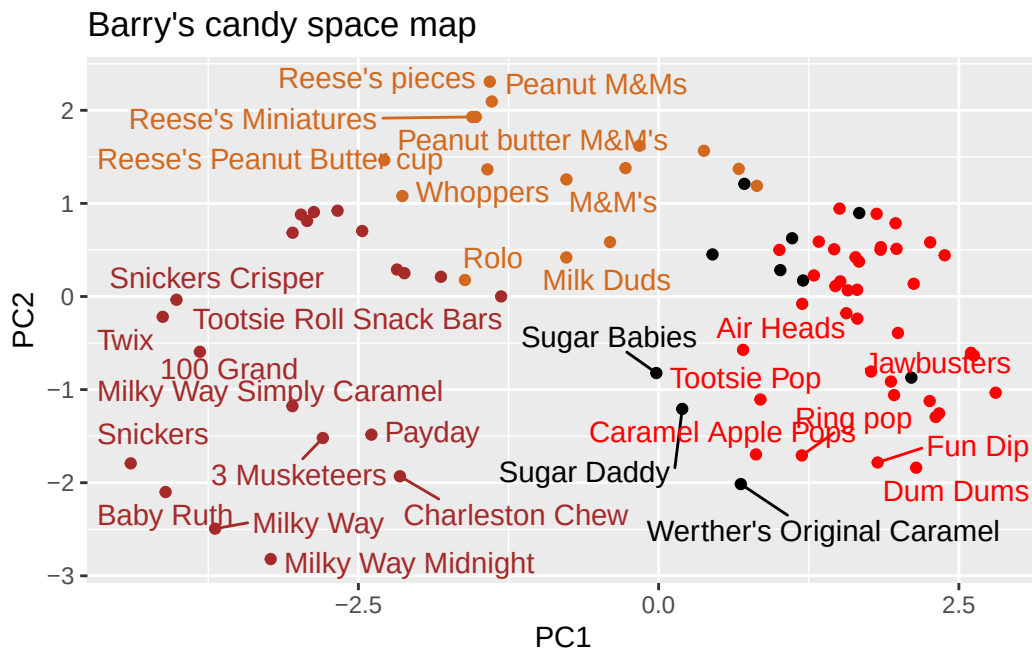
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

Let's look at our first main result figure - the "PC plot" or PC1 vs PC2

```
ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols) +
  labs(title="Barry's candy space map")
```

Warning: ggrepel: 54 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Don't forget about your variable "loadings" - how the original variables contribute to your new PCs...

```
ggplot(pca$rotation) +
  aes(PC1, rownames(pca$rotation)) +
  geom_col()
```

